

12/11/14 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)
 Semester: July 2014 - November 2014

Max. Marks: 100**Duration: 3 hrs.****End Semester Exam**

Class: SE

Semester: III

Branch: COMP

Course code : **UCEC 305**

Name of the Course: Discrete Structures

Instructions:**1. All Questions are Compulsory**

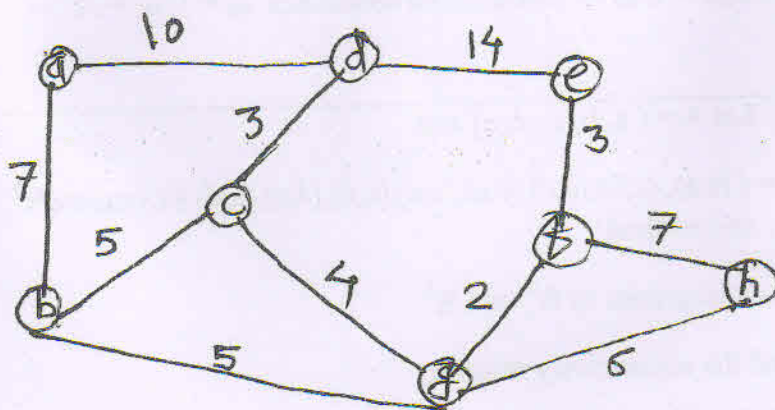
Question No.		Max. Marks
Q 1	a) Among the integers 1 and 300 (i) How many of them are not divisible by 3 nor by 5 nor by 7? (ii) How many of them are divisible by 3 but not by 5 nor by 7?	5
	b) Show that if 30 dictionaries in a library contain a total of 61,327 pages, then one of the dictionary must have at least 2045 pages.	5
	c) Prove that $8^n - 3^n$ is a multiple of 5 by mathematical induction.	5
	d) Solve the recurrence relation $a_{n+2} - 5a_{n+1} + 6a_n = 2$ with initial conditions $a_0 = 1, a_1 = -1$.	5
Q 2	a) Let $A = (a, b, c, d, e)$ and $R = ((a,a), (a,b), (a,c), (b,c), (c,e), (c,d), (d,e), (a,d))$ Compute R^2, R^3 (by any method) Draw diagraphs of R^2 and R^3. Find the connectivity relation.	10
	b) Let $A = Z$, and let R be defined as aRb iff $a \leq b$. Is R an equivalence relation.	5
	c) Show that using Laws of Logic $[(P \rightarrow Q) \wedge \sim Q] \rightarrow \sim P$ is a Tautology	5

Q 3	a) Construct the Hasse diagram of D_{42} . Is it a Lattice ? If yes , check the distributive property.	10
	b) Let $f(x) = x^3$, $g(x) = 4x^2 + 1$, $h(x) = 7x - 2$. Find $f \circ (h \circ g)$ and $g \circ (h \circ f)$. Is it associative.	05
	c. Let $A = \{1, 2, 3, 4, 5\}$ and the relation $R = \{(1, 1), (1, 4), (2, 2), (3, 4), (3, 5), (4, 1), (5, 2), (5, 5)\}$ Find the transitive closure of R by Warshall's Algorithm.	05

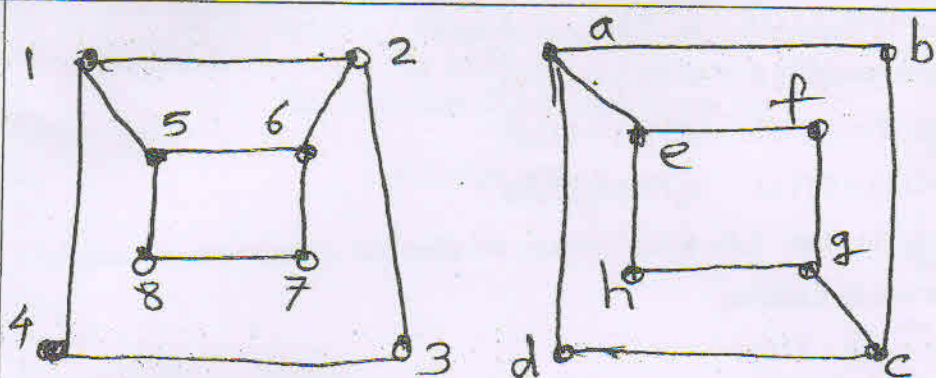
or

Q 3	a) Let $A = (1, 2, 3, 4)$, R is a relation defined as aRb iff $b \leq a+1$. Determine whether R is reflexive, irreflexive, symmetric, asymmetric, antisymmetric ?	5
	b) Draw the Hasse diagram for the following relation as aRb iff $a b$. $A = (1, 2, 3, 4, 12)$	5
	c) Find the generating function for the following sequence $1, 2, 3, 4, 5, 6, \dots$	5
	d) If $f : A \rightarrow B$ be both one to one and onto then prove that f^{-1} is also both one to one and onto.	5

Q 4	a) Find Minimum Spanning Tree for the following graph using Kruskals algorithm.	5
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	b) Determine whether the following graphs are isomorphic ?	5
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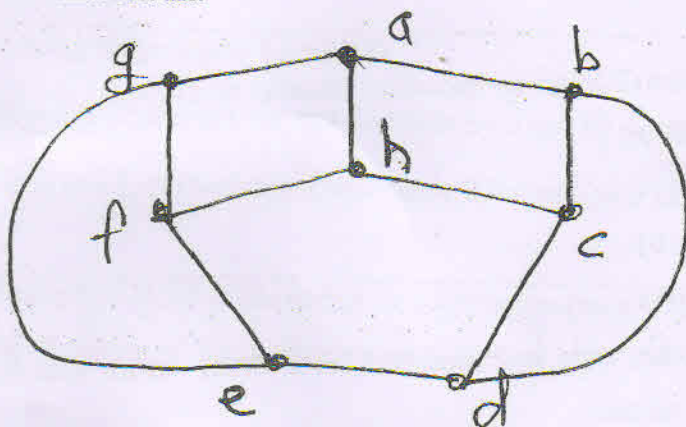


c) How many edges must a planer graph have if it has 7 regions and 5 nodes

5

d) Define Hamiltonian path and circuit. Find Hamiltonian path and circuit.

5



Q 5

a) Let $H =$

100

011

111

100

010

001

Be a parity check matrix, Determine the group code $e_H : B^3 \rightarrow B^6$

10

b) Show that $(2, 5)$ encoding function $e : B^2 \rightarrow B^5$ defined by

$e(00) = 00000$ $e(01) = 01110$ $e(10) = 10101$

$e(11) = 11011$ is a group code.

05

c) Prove that the set $A = (0, 1, 2, 3, 4, 5)$ is a group under addition modulo 6.

05

or

Q 5

a) Consider the $(3, 5)$ group encoding function $e : B^3 \rightarrow B^5$ as

10

	$e(000) = 00000$ $e(100) = 10011$ $e(001) = 00110$ $e(101) = 10101$ $e(010) = 01001$ $e(110) = 11010$ $e(011) = 01111$ $e(111) = 11100$ Decode the following words relative to maximum likelihood decoding function. 1. 11001 2. 01010 3. 00111	
	b) Prove that the set $A = (1, 2, 3, 4, 5, 6)$ is an abelian group under multiplication modulo 7.	5
	c) Consider the $(2,5)$ group encoding function $e : B^2 \rightarrow B^5$ as $e(00) = 00000$ $e(01) = 01110$ $e(10) = 10101$ $e(11) = 11011$ Construct the table and find out How many errors will it detect and correct.	5